

Kaung Nyo Lwin

AI Engineer & Researcher

Pathum Thani, Thailand

kaungnyolwin@gmail.com

+66 64 832 8635

kaung-nyo-lwin.github.io

GitHub / Kaung-Nyo-Lwin | LinkedIn

AI engineer building computer vision and generative AI applications, from model training and evaluation to cloud deployment. Experience spans retrieval-augmented generation, LLM fine-tuning, biometric verification, and intelligent video analytics, backed by Python engineering and solution architecture.

GENERATIVE AI	COMPUTER VISION	ML ENGINEERING & CLOUD
LLM fine-tuning, LoRA, GRPO, RAG, embeddings, semantic search; Hugging Face Transformers, TRL, LangChain, LlamaIndex, OpenAI API, FAISS.	PyTorch, OpenCV, YOLO, ByteTrack, ONNX, TensorFlow Lite; object detection, multi-object tracking, biometrics, model evaluation.	Python, SQL, FastAPI, Docker, Google Cloud, BigQuery, Vertex AI, Terraform, PostgreSQL, Git; data pipelines and cloud deployment.

AI & Machine Learning Experience

Jul 2026–Present

CV Researcher

AIT × Modular Open Source Identity Platform (MOSIP)

Biometric AI · Face liveness and anti-spoofing

- Develop deep learning workflows for face anti-spoofing and liveness detection, supporting biometric verification for MOSIP digital identity.
- Engineer PyTorch training pipelines and model evaluation across ONNX and TensorFlow Lite, with demographic analysis and export for mobile SDK integration.

Jul 2026–Present

CV Engineer

AIT × CDG Group

Driver Licensing Management System

- Engineer a computer vision system for driving-test assessment, turning camera footage into calibrated vehicle trajectories and reviewable test evidence.
- Build OpenCV pipelines for camera calibration, perspective correction, and marker tracking, with RTSP capture and browser tools for field setup and validation.

Mar 2026–Present

Research Engineer

AIT × Toyota Mobility Foundation

TRUST · Intelligent transport and road safety

- Build a cloud analytics platform that combines vehicle telemetry and CCTV analysis to help researchers investigate traffic behavior and road-safety risks.
- Develop object detection and multi-object tracking pipelines with YOLO and ByteTrack, connecting vehicle trajectories to geospatial analysis.
- Deploy FastAPI services and BigQuery analytics on Google Cloud using Docker and Terraform, delivering authenticated dashboards for research teams.

Jul 2025–Oct 2025

LLM Developer Intern

AI Brain Lab

Generative AI · Document intelligence

- Developed generative AI applications for ESG reporting and financial-document question answering using LLMs, retrieval-augmented generation (RAG), and FastAPI.
- Built document ingestion, table extraction, and embedding pipelines for PDF, Excel, and CSV sources, enabling semantic search across Stock Exchange of Thailand materials.

Research

Completed
master's thesis

L2BC | Learning with Verifiable Rewards
Asian Institute of Technology

- Developed a reinforcement learning framework using GRPO and LoRA to train compact LLMs for text-to-visualization with rule-based, verifiable rewards.
- Built verifiers that execute generated code, reconstruct chart type and data, and check correctness against ground truth; used this feedback to adapt training.
- Demonstrated single-pass generation competitive with larger agentic pipelines on VisEval, with ablation and generalization studies to identify what the verifier teaches.

Solution Architecture & Earlier Experience

Aug 2026–Present

Solutions Architect (Contract)
MA'HAR EMPEROR (Audit Firm)
Project Management & Tracking System

- Own end-to-end delivery of a project management system, from requirements and data collection through solution architecture, development, and production rollout.
- Build Odoo/PostgreSQL workflows for project planning, timesheets, access control, and management reporting, translating audit operations into a maintainable business system.

Feb 2023–Jul 2024

Senior Business Analyst
KEYfields

- Translated accounting and logistics requirements into SQL data models, stored procedures, and reporting workflows for SaaS products.
- Automated deployments and database backups with Python and PowerShell, improving operational reliability and reducing manual effort.

May–Sep 2020

Senior Asset Management Analyst
Telecom International Myanmar
Previously Asset Management Analyst · Jun 2017–Apr 2020

- Built Excel/VBA reporting tools and KPI dashboards to streamline asset tracking, inventory planning, and telecom rollout coordination.

Feb–May 2016

Telecom Installation Engineer
Ericsson Myanmar

- Delivered site surveys, equipment configuration, and network testing for telecom installations.

Education

Master of Engineering
Data Science and Artificial Intelligence
Asian Institute of Technology, Thailand
2024–2026

Bachelor of Engineering
Avionics
Myanmar Aerospace Engineering University
Myanmar · 2010–2015

Certifications & Skills

IBM: Data Science, Data Analyst. DeepLearning.AI: Machine Learning, Deep Learning, TensorFlow Developer, Mathematics for Machine Learning and Data Science. University of Illinois: Fundamentals of Accounting.

Additional tools scikit-learn, NumPy, pandas, MLflow, Power BI, SQL Server, PowerShell, Excel/VBA, Odoo.

Languages Myanmar (native) · English (C1)